

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 1**

**COURSE NAME:** Cryptography and Network  
Security **COURSE CODE:** CSA5116

### **COURSE OUTCOME COVERED**

**CO1:** Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

**QUESTIONS: 5**

**Total Marks:50**

Annexure A

### **APPLICATION- BASED QUESTIONS**

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**Code of Conduct:**

(192324202)

I K. Harshitha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K. Harshitha

**Signature of the student:**

## APPLICATION- BASED QUESTIONS

### 1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

### 2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- Deduce the most likely original message.
- Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

### 3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- Demonstrate how the encryption process transforms the message step-by-step.
- Explain how the repeating keyword influences each character of the ciphertext.

### 4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- Analyze how investigators could detect the presence of hidden data within the image.
- Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

### 5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAVDEN".

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

# **RUBRICS FOR CALCULATION:**

| Criteria                  | Weightage | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                              |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25        | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30        | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20        | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |
| Analysis of Results       | 15        | Insightful analysis with accurate interpretation and validation of results                    | Good analysis with reasonable interpretation                | Basic analysis with limited insights                | Poor or incorrect analysis of results            |
| Documentation             | 10        | Clear, well-structured documentation with diagrams, code, and results                         | Organized documentation with necessary details              | Basic documentation with missing details            | Poor or incomplete documentation                 |

## Scenario :- Secure Messaging System :-

A company deploys a Keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a) Demonstrate how the encryption process transforms the message step-by-step

b) Explain how the encryption repeating keyword influences each character of the ciphertext.

⇒ A company uses a keyword-based encryption technique for internal communication.

- Plaintext : HELLO

- Keyword : KEY

This is typically demonstrated using the Vigenere Cipher, where each letter of keyword is combined with the corresponding letter of the message.

a) Encryption Message :- (step-by-step)

Step 1 :- write the plaintext.

The employee wants to send:

HELLO

Plain text :-     H       E       L       L       O

Position :-       1       2       3       4       5

Step 2 :- Repeat the keyword

The keyword: K E Y K E

KEY :-

Position :- 1 2 3 4 5

Plain text :- H E L L O

keyword :- K E Y K E

NOTICE :- The key starts again from K

Step 3 :- convert letters into numerical values

In the Vigenere cipher, we normally assign:

A=0 ..... Z=25

|        |   |   |    |    |    |
|--------|---|---|----|----|----|
| Letter | H | E | L  | L  | O  |
| value  | 7 | 4 | 11 | 11 | 14 |

For the keyword

|        |    |   |    |    |   |
|--------|----|---|----|----|---|
| Letter | K  | E | Y  | K  | E |
| value  | 10 | 4 | 24 | 10 | 4 |

Step 4 :- Apply the encryption formula

The Vigenere encryption formula is:

$$C = (P + K) \text{ mod } 26$$

where:-

C = ciphertext value

K = keyword value

P = Plaintext value

mod 26

character 1:-  $H+K \rightarrow (7+10) \bmod 26 \rightarrow 17 \rightarrow H \rightarrow R$

$(E+E) \rightarrow (4+4) \bmod 26 \rightarrow 8 \rightarrow I$

character 3:-  $L+Y \rightarrow (11+24) \bmod 26 \rightarrow 9 \rightarrow J$

character 4:-  $L+K \rightarrow (11+10) \bmod 26 \rightarrow 21 \rightarrow V$

character 5:-  $O+E \rightarrow (14+4) \bmod 26 \rightarrow 18 \rightarrow S$

Therefore:-

HELLO  $\rightarrow$  RIJVS

So, the employee sends the ciphertext:-

RIJVS

Instead of original word HELLO

Simple Representation :-

**P. Text** :- H E L L O

**Keyword** :- K E Y K E

**Ciphertext** :- R I J V S

The receiver, knowing the keyword KEY, can reverse the process using:-

$$P = (C - K) \bmod 26$$

and recover HELLO.

b) Repeating keyword influences each character

The keyword determines how much each plaintext character is shifted.

Here, the keyword is :-

KEY

Its numerical values are:

•  $K=10$

•  $E=4$

•  $Y=24$

For 5-characters

HELLO

KEYKE

Plaintext

Keyword

Shift

H

K

10

E

E

4

L

Y

24

L

K

10

O

E

4

this means the same plaintext letter that can produce different ciphertext letters

First L:-

$$L + Y = J$$

Second L:-

$$L + K = V$$

L → J

L → V

This is an important property of the Vigenere cipher

Why is the repeating keyword useful?

In a simple Caesar cipher, every character is shifted by the same amount.

HELLO → KHOOR

But in a keyword-based cipher

Plaintext :- H E L L O

Keyword :- K E Y K E

Ciphertext :- R I J V S

Different keyword characters produce different shifts

Therefore, the cipher text is less obvious than a simple substitution with one fixed shift.

Important points :-

The keyword does not directly replace the plaintext characters.

Instead, each keyword character provides a numerical shift that is added to the corresponding plaintext character.

SO, Ciphertext = (Plaintext + Repeating Key)

#### 4) Scenario :- Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a) Analyze how investigators could detect the presence of hidden data within the image.
- b) Recommend two techniques that could strengthen the security of such hidden communication against detection (or) extraction.

During a digital forensic investigation, investigators suspect that a company logo contains hidden confidential information.

→ The image looks completely normal when opened normally.

→ This is classic example of steganography.

⊛ What is steganography?

Steganography is the technique of hiding information inside another ordinary-looking file such as:

- Image
- Video
- Audio
- Document.

The important idea is that the existence of the secret message itself is hidden.

For example :-

Normal image



Hidden confidential information



Image appears normal to ordinary users

→ This differs from encryption.

Encryption vs. steganography

Encryption:-

HELLO



Encrypted text



RIJVS

The message is clearly transformed and may look suspicious.

Steganography:-

Secret message + normal image



Modified image →

Looks like normal image

a) A forensic investigation can proceed in several stages

① Preserve the original image

The first step is to make a forensic copy of the suspicious file.

Investigators should :-

- Preserve the original file
- Work on a copy
- Calculate a hash value such as SHA-256
- Record file size, creation time, modification time and other metadata.

this ensures that the investigation does not accidentally modify the evidence.

② Examine image metadata

Investigators can inspect metadata associated

- File type
- Image dimensions
- creation data.

Image:- compay-logo.jpg

software:- unknown

comment:- unusual encoded text.

Two techniques that could strengthen hidden communications.

If the question is asking from the security perspective, two techniques can be recommended.

Technique 1:- Encrypt the secret message before hiding it.

→ Instead of directly hiding readable confidential information inside the image:

Confidential message



Steganography



Image

the sender can first encrypt the message:

Confidential message



Encryption



Cipher text



Steganography



Image

Why is this stronger?

Suppose investigators discover the hidden data.

→ The attacker should detect

- hidden information
- Decrypt the extracted information.

Technique 2: use a robust and well-designed hiding method.

A simple hiding method may place secret bits in an predictable sequence:

Pixel 1  $\rightarrow$  secret bit

Pixel 2  $\rightarrow$  secret bit

Pixel 3  $\rightarrow$  secret bit

...

$\rightarrow$  This can create statistical patterns that specialized steganalysis tools may detect.

A stronger approach can use a secret key

Secret message  $\downarrow$

secret embedding key

$\downarrow$

Pseudo-random locations

$\downarrow$

Image

$\rightarrow$  only someone who possesses the correct key knows where the hidden information has been placed.

This makes extraction more difficult because an investigator cannot simply assume that every pixel (&) every least-significant bit contains the secret message.